

Client p , $p = 1, \dots, P$ (the data X_p never leave the client)

A1: PRE mass A2: gate

A1: POST mass A3: offsets

(1) At the broadcast V

memberships $\bar{u}_{pkj}$
PRE mass $\bar{M}_{pj}$
gates: r_{pj} (footprint),
 q_{pj} (mass)

(2) L local FCM steps

from v_j (or $v_j + \delta_{pj}$)
local centres c_{pj}
POST mass M_{pj}^L
offsets δ_{pj} updated

(3) Upload

local centres c_{pj}
with Mass $_{pj}$ and Gate $_{pj}$

broadcast V

Server

prototypes
 $V = (v_1, \dots, v_c)$

$V \leftarrow V^+$, next round

Server update

$$w_{pj} = \beta_p \text{Mass}_{pj} \text{Gate}_{pj},$$

$$v_j^+ = (1 - \alpha_s) v_j + \alpha_s \frac{\sum_p w_{pj} c_{pj}}{\sum_p w_{pj}}$$

upload

Design axes. A1 mass point: PRE ($\bar{M}_{pj}$) or POST (M_{pj}^L). A2 gate: none, footprint r_{pj} or mass q_{pj} (two support masks serve as probes). A3 personalization: offsets δ_{pj} off or on. **Analysed for every rule:** the objective it minimizes, if any, and its fixed points; the tilt geometry of its weight; its held-out accuracy and objective; its regret across configurations.